PLANT PROPAGATION · TEAM PROJECT

Plug Lab

Choose your propagation adventure

Your team is starting a propagation nursery that grows **plug trays**: a 1020 tray with a cell insert, sold to other growers. You'll choose where your plants come from, research every cost, design your benches, set a price, and compare your business with the other teams.

1. Open **Plug Lab** and go to the **Lab** tab. Work through the steps in order.
2. Each step tells you which section of this worksheet to fill in. Write your numbers here first, then enter them in the app.
3. Write down **where every number came from**: a website, a person, a price list, or your own measurement.
4. Save your results file in the last step, turn it in, and present to the class.

App link _____

1 Your team

In the app: Lab step 1

Team name _____

Role	Name	What you own
Researcher		
Grower		
Bookkeeper		
Presenter		

Researcher: finds prices and cites sources. Grower: growing plan, time trials, benches. Bookkeeper: enters numbers in the app and checks the math. Presenter: builds the slides and leads the talk. Everyone helps with everything; your role is the part you're responsible for.

2 Choose your adventure

In the app: Lab step 2

Where will your plants come from? Each path sends you to different research. Check one.

☐ **A Grow your own mother stock**

Pick a plant at the college nursery, keep stock plants on a bench, and take cuttings from them again and again.

You'll research: nursery plant prices, how many cuttings one plant gives, stock bench space.

☐ **B Buy in cuttings**

Order unrooted cuttings from a propagator such as North Carolina Farms (ncfarmsinc.com).

You'll research: a variety, price per cutting, minimum order, shipping, patents.

☐ **C Collect cuttings offsite**

Drive to a garden or growing site and take cuttings, with the owner's permission.

You'll research: mileage, your car's mpg, gas prices (AAA), the IRS mileage rate, collecting time.

☐ **D Grow from seed**

Buy seed and sow it right in the plug tray, for example from the Theodore Payne Foundation.

You'll research: seed price, seeds per packet, germination time, seeds per cell.

Why did your team choose this path?

What do you predict will be your biggest cost? Why?

> Fill in **only your path's page** in section 3 (3A, 3B, 3C or 3D). Skip the other three.

3A Path A · Mother stock

In the app: Lab steps 3 and 5

Walk the college nursery and choose a plant you could take cuttings from again and again.

Botanical name and cultivar _____
Example: Salvia 'Hot Lips'

Container it's in ◯ #1 ◯ #3 ◯ #5 ◯ Other _____

Price of one plant _____ $
From the nursery price list

Source .....

PP or PPAF number on the tag _____
Write "none" if it isn't patented. Patented plants need a license to propagate.

Royalty per plug, if patented _____ $
Ask your instructor how to estimate it

Count the cuttings

Shoots you could cut from one plant today _____

Weeks for shoots to grow back _____ weeks

Cuttings per plant per month	shoots ÷ (weeks to grow back ÷ 4.3)

Months you will keep cutting from one plant _____ months

Upkeep per plant per month	water + fertilizer + care for one plant, one month. Show your math.

Stock pot size for your stock bench ◯ #1 ◯ #3 ◯ #5

3B Path B · Buying in cuttings

In the app: Lab steps 3 and 5

Browse North Carolina Farms' unrooted cuttings and choose a variety.

Botanical name and cultivar _____

Supplier and product page _____

Price per cutting _____ $

Source .....

Minimum order _____ cuttings

Cuttings you would order at a time _____ cuttings

Freight (shipping) per order _____ $

Unrooted cuttings ship fast in a cooled box. Explain your estimate below.

How did you estimate freight?

_____

_____

PP or PPAF number _____

Write "none" if it isn't patented

Does the price include the royalty? ◯ Yes ◯ No ◯ Not sure

Cost per cutting with freight	price per cutting + freight ÷ cuttings per order

Cutting cost per plug tray	cells × cuttings per cell × cost per cutting

3C Path C · Collecting offsite

In the app: Lab steps 3 and 5

Choose a site you can reach and get permission first. Never take cuttings from parks or wild land without a permit.

Botanical name and cultivar _____

Is it patented? ◯ No ◯ Yes: pick a different plant

Collecting site _____

Who gave permission, and when _____

Trip cost

Round-trip miles _____ miles
From a map app

Source .....

Your car's miles per gallon _____ mpg

Gas price, and the date you checked _____ $/gal
AAA gas prices for California

Source .....

Gas cost per trip	round-trip miles ÷ mpg × gas price

IRS standard mileage rate _____ $/mile
irs.gov · covers gas and wear on the car

Source .....

IRS cost per trip	round-trip miles × IRS rate

Which trip cost will you use? ◯ Gas only ◯ IRS rate

Why?

_____

Cuttings you can take per trip _____ cuttings
How many you can take and still stick them fresh

3D Path D · Growing from seed

In the app: Lab steps 3 and 5

Browse the Theodore Payne seed store or Calscape and choose a California native that grows easily from seed.

Botanical name (and common name) _____

Supplier and product page _____

Price per packet (or ounce or pound) _____ $

Source .....

Seeds in that amount _____ seeds

Cost per seed	price ÷ seeds

Shipping per order _____ $

Write 0 if you would pick it up

Germination

Pretreatment needed? _____

Soaking, cold stratification, smoke, scarification, or none

Time to germinate _____ weeks

Source .....

Best temperature to germinate _____ °F

Seeds per cell _____ seeds

Why that many seeds per cell?

Seed cost per plug tray	cells × seeds per cell × cost per seed

4 Plan how it grows

In the app: Lab step 4

Research how your plant is propagated. Use your textbook and lecture notes, the supplier's growing information, and Calscape for California natives.

Cells per plug tray ○ 32 ○ 38 ○ 50 ○ 72 ○ 128

Why that size?

Cuttings or seeds per cell _____

Weeks under mist (Path D: weeks to germinate) _____ weeks

Source .....

Weeks hardening off (Path D: weeks growing on) _____ weeks

Source .....

Rooting % (Path D: % of cells that fill) _____ %

Source .....

Grade-out loss: rooted but not good enough to sell _____ %

Rooting hormone? ○ Yes ○ No ○ Path D

If yes, which product and strength _____

Sellable plugs per plug tray	cells × rooting % × (1 – grade-out %)

What could go wrong with this crop, and how would you prevent it?

5 Time the work

In the app: Lab step 6

Do each job in the lab for a few minutes and count. Do it three times and use the average.

Wage

Source	Job or wage it lists	Wage per hour

Wage you will pay _____ $/hr

Time trials

Job	Trial 1 count · seconds	Trial 2 count · seconds	Trial 3 count · seconds	Average
Taking cuttings (A, C)				
Sticking cuttings or sowing cells				
Filling a plug tray with mix				
Grading and packing a plug tray				

Per hour = count ÷ seconds × 3,600 · Minutes per plug tray = seconds ÷ 60

What slowed you down? How could a real nursery go faster?

_____

_____

6 Potting mix and plug trays

In the app: Lab step 7

Propagation mix product _____

Price per cubic yard _____ $/yd³

Only have a bag price? Price ÷ cubic feet in the bag × 27

Source .....

Packing and waste _____ %

Extra mix lost to packing cells and spills

Plug tray and insert product _____

Price for one tray with insert _____ $

Source .....

Times you can reuse it _____ times

Cell volume from the spec sheet _____ in³

7 Supplies

In the app: Lab step 8

What does a propagation house buy over and over?

Supply	Amount you buy	Price	Bought every	Our plant uses it?	Source
Rooting hormone					
Fungicide					
Fertilizer					
Plant tags					
Sticky cards					

8 Design your benches

In the app: Lab step 9

Start with the mist bench. Then size the open bench so every round has somewhere to harden off.

Mist bench _____ ft
length × width × how many

Plug trays it holds (app) _____ trays

Open bench _____ ft
length × width × how many

Plug trays it holds (app) _____ trays

Open bench needed	mist trays × (hardening weeks ÷ mist weeks, round up)

Balanced? ◯ Yes ◯ No

Path A: stock bench (app) _____ ft

Sketch your layout. One square = 1 foot. Draw each bench to scale with 2-foot aisles. Label sizes and bench types.

9 Greenhouse costs

In the app: Lab step 10

A greenhouse costs money whether it's full or empty. Estimate each cost for one year.

Cost	Per month	Per year	How you estimated it, and source
Structure payment or depreciation			
Utilities (power, gas)			
Water			
Insurance			
Shared labor (watering, spraying)			
Other overhead			

% of bench in use _____ %

Benches are never 100% full. What's realistic?

Weeks a year you grow _____ weeks

Mist add-on per sq ft per week _____ $

Extra cost of mist and bottom heat. 0 to leave it out.

Bench rate the app figured _____ $/sq ft/week

Which greenhouse cost was hardest to estimate? Why?

_____

_____

10 Set your price

In the app: Lab step 11

Cost per plug (app, Price tab) _____ $
This is break-even

Target margin _____ %

Suggested price at that margin (app) _____ $

Seller	Plant and size	Price per liner or plug	Source

Our price per plug _____ $

Why this price? Can you compete and still make a profit?

_____

_____

11 Contract challenge

In the app: Lab step 12

Your instructor will give you a buyer's order.

Plug trays ordered _____ **Fits on your benches?** ◯ Yes ◯ No

Ship date _____ **First stick or sow date** _____

Plugs to deliver (app) _____ **Profit on the order** _____ $

If it doesn't fit, what would you change?

_____

_____

12 Your results

In the app: Lab step 13

Copy these from the app when every step is done.

Cost per plug tray _____ $

Sellable plugs per tray _____

Cost per plug _____ $

Price per plug _____ $

Margin _____ %

Plug trays per year _____

Profit per year _____ $

Bench area _____ sq ft

Where the money goes (per plug tray)

Cost	$ per plug tray	% of total
Cuttings or seed		
Potting mix		
Plug tray		
Supplies		
Labor		
Bench space		
Royalty		

Our biggest cost _____

One way to make it smaller:

_____

_____

☐ Results file saved and turned in ☐ Blueprint JPG exported ☐ Price list PDF exported

13 Compare with the class

In the app: Class board

Fill this in during the presentations or from the class board.

Team	Path	Plant	Cost / plug	Price	Profit / yr	Something they did differently

1. Which path had the lowest cost per plug? Why do you think so?

2. Which cost was the biggest for most teams?

3. If your team started over, which path would you choose? Why?

4. Name one way your team could raise profit without raising your price.

5. What number surprised you the most?

14 Plan your presentation

In the app: Share your results

About 5 minutes. Every team member speaks.

Slide	What to show	Who presents
1		
2		
3		
4		
5		
6		
7		
8		

Suggested slides: 1 team and plant (with a photo) · 2 your path and how you researched it · 3 growing plan · 4 bench blueprint · 5 where the money goes · 6 price and profit · 7 contract challenge · 8 what you learned.

How you'll be scored

Part	Points	What it looks like
Research and sources	20	Every number has a source or a measurement behind it.
Accurate numbers	20	Numbers match between the worksheet and the app; math is shown.
Bench design	15	Benches are balanced and drawn to scale.
Price decision	15	Price is backed by your costs and real market prices.
Presentation	20	Clear, on time, everyone speaks, slides use your blueprint and results.
Worksheet and teamwork	10	Worksheet complete; each role done.
Total	100	